Sometimes a theorem states that several propositions are equivalent. Such a theorem states that propositions $p_1, p_2, \ldots, p_n$ are equivalent. This can be written as $p_1 \leftrightarrow p_2 \leftrightarrow \ldots \leftrightarrow p_n$, which states that all n propositions have the same truth values, and consequently, that for all i and j with $1 \le i \le n$ and $1 \le j \le n$, p_i and p_j are equivalent. One way to prove these are mutually equivalent is to use the tautology

$$\boxed{p_1 \leftrightarrow p_2 \leftrightarrow \ldots \leftrightarrow p_n \leftrightarrow (p_1 \to p_2) \wedge (p_2 \to p_3) \wedge \ldots \wedge (p_n \to p_1)}.$$

This shows that if the n conditional statements $p_1 \to p_2, p_2 \to p_3, \ldots, p_n \to p_1$ can be shown to be true, then the propositions $p_1, p_2, \ldots, p_n$ are all equivalent. This is much more efficient than proving that $p_i \to p_j$ for all $i \ne j$ with $1 \le i \le n$ and $1 \le j \le n$. (Note that there are $n^2 - n$ such conditional statements.)

When we prove that a group of statements are equivalent, we can establish any chain of conditional statements we choose as long as it is possible to work through the chain to go from any one of these statements to any other statement. For example, we can show that p_1, p_2 and p_3 are equivalent by showing that $p_1 \to p_3$, $p_3 \to p_2$ and $p_2 \to p_1$.

Example 14: Show that these statements about the integer n are equivalent:

p_1 : n is even p_2 : $n-1$ is odd p_3 : n^2 is even

Solution: We will show that these three statements are equivalent by proving that each of $p_1 \to p_2$, $p_2 \to p_3$ and $p_3 \to p_1$ are ~~equivalent~~ True.

Case 1: $p_1 \to p_2$. We will give a direct proof. Let p_1 be True

$\therefore$ Let n be an arbitrary even integer. By definition 1, $\exists$ an integer K such that $n = 2K$. $\therefore n-1 = 2K-1 = \boxed{2(K-1)+1}$. $\because K$ is an integer, $K-1$ is an integer. By defn. 1, $(n-1)$ is an odd integer. $\boxed{\therefore p_2 \text{ is True}}$